

Project Synopsis

TITLE : A Semantic Search Engine Using Rust

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1) Motivation and Background

Keyword-based search (inverted indices) is fast and proven, but it often fails when the user's intent is semantic rather than lexical, especially for technical, academic, or conceptual queries where the best source may not share the same keywords. At the same time, LLMs can generate fluent answers but frequently struggle with grounding. Users cannot reliably trace a statement back to the exact original source fragment.

This project addresses both problems by building a neural retrieval engine. Instead of generating answers, it locates high-signal evidence using dense embeddings and returns the exact text span and URL for verification.

2) Problem Statement

Modern information access faces two systemic failures:

1. **SEO-driven noise in web search**
Many top results are optimized for ranking rather than information density, pushing primary sources and technical references down the stack.
2. **Hallucination and weak attribution in generative systems**
Generative AI can summarize, but may misstate details. Even when correct, it often lacks direct, verifiable citations to the original context.

3) Objectives

This project delivers a retrieval-first system with four concrete goals:

- **Asynchronous ingestion at scale**
Build a high-concurrency, fault-tolerant crawling and preprocessing pipeline in Rust.
- **Semantic vectorization of unstructured text**
Convert documents into dense embeddings (for example, 384 or 768 dimensions) using Transformer encoders, with chunking that preserves local context.
- **Low-latency Approximate Nearest Neighbor (ANN) retrieval**
Implement an ANN index (HNSW) capable of sub-second similarity search over large corpora.
- **Direct source mapping for full verifiability**
Maintain a strict 1:1 mapping from each vector to the original source fragment (paragraph or span) plus URL so every result is traceable and auditable.

4) Methodology

Data Preparation

Curate a high-signal corpus (academic papers, documentation, public archives, verified wikis) to reduce spam and SEO bias and enable meaningful evaluation.

Pipeline Optimization

Optimize end-to-end efficiency by overlapping network I/O, parsing, embedding, and indexing. The goal is to keep CPU and GPU utilization stable and minimize idle time in any stage of the pipeline.

5) Expected Outcome and Key Advantages

This project produces a search utility that excels in two areas:

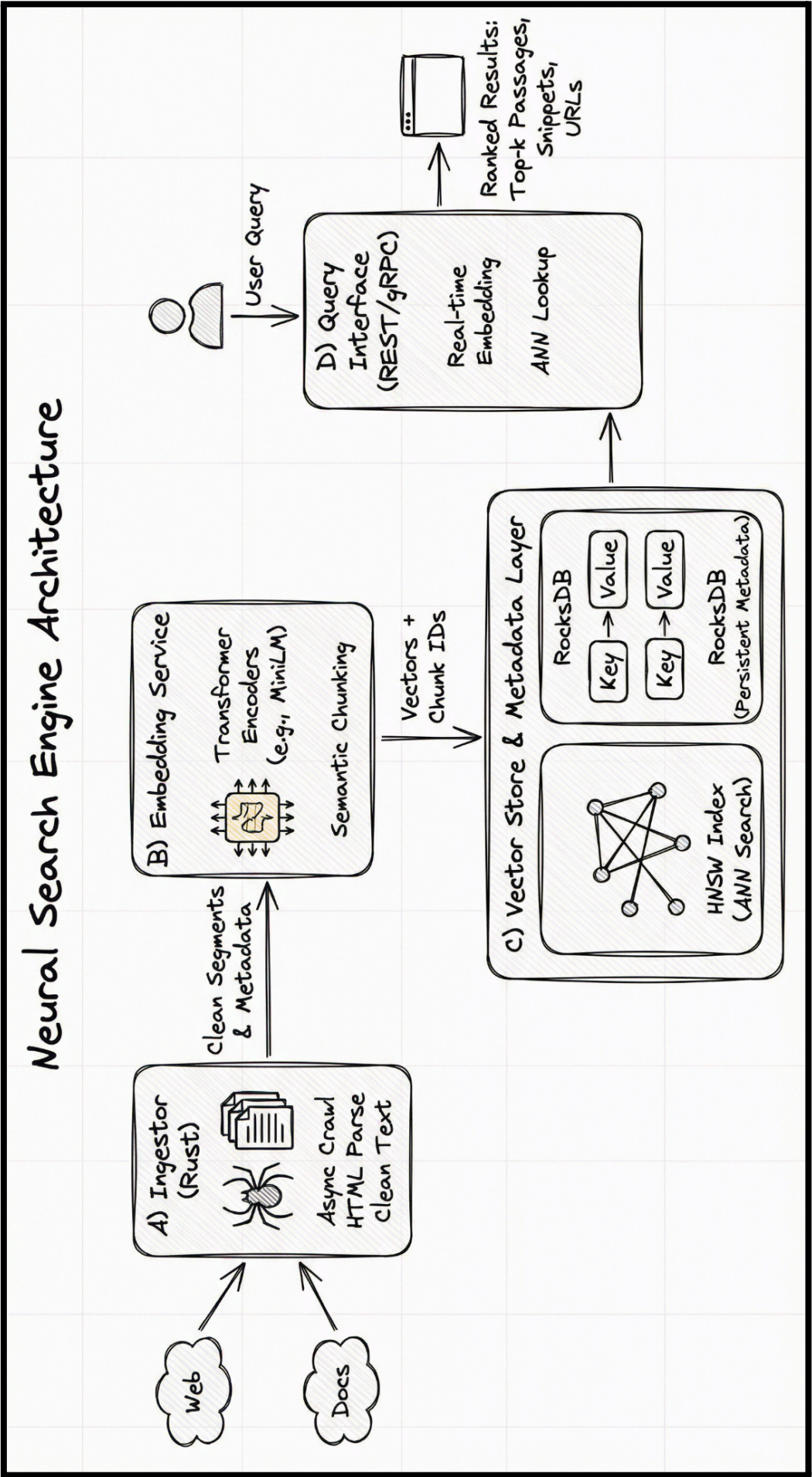
Better than traditional web search for semantic intent

Instead of matching keywords, the engine retrieves results by meaning, surfacing relevant sources even when terminology differs. This is especially valuable for technical and research queries.

More reliable than generative systems for grounding

The system does not answer by synthesis. It retrieves evidence. Each result is a directly citable passage with the exact URL and context, removing hallucination at the retrieval layer.

6) Architecture



7) Foundational Difference

